

ROOT CAUSE ANALYSIS



DATE

HOSPITAL NUMBER

CHAROENKRUNG PRACHARAK
HOSPITAL

8 Charoenkrung rd.
Bangkholeam, Bangkholeam
BKK, Thailand
10120
02-289-7000

PATIENT PROFILE

Thai female

25 years old

BMI 37 kg/m²

Employee

Buddhist

Social Security Rights Bangpakok 8 Hospital

**G₁ GA 37⁺¹ weeks (by LMP) with
GDMA1 with obesity**

7 hours PTA labour pain with mucous bloody show, no ruptured of membrane

DATE	TIME	EVENT	DESCRIPTION
Oct 21	13.10	Admission	V/S BT 36.0 C, P 94 bpm, R 20/min, BP 137/68 mmHg PV: Cx 2 cm, 50%, -2, ML FHR 140 bpm, UC 5', 40", 1+
	13.19-14.40	COVID-19	13.19 Nasal swab for COVID-19 14.40 Specimen was taken at Lab
	14.50	Augmentation	LRS 1,000 mL + oxytocin 10 units IV drip titrate until good UC 5% D/N/2 1,000 mL IV drip 60 mL/hr
	15.17	COVID-19	COVID-19 PCR rapid test was reported COVID-19 PCR rapid test: undetectable
	16.00		FHR 150 bpm, UC 5', 40", 2+ On oxytocin drip rate 10-20 mL/hr
	17.00		PV: Cx 2 cm, 75%, -1, MR FHR 140 bpm, UC 4', 45", 2+

DATE	TIME	EVENT	DESCRIPTION
			On oxytocin drip rate 30-40 mL/hr
	18.00		V/S BT 36.2 C, P 90 bpm, R 22/min, BP 141/78 mmHg FHR 140 bpm, UC 2'40", 45", 2+ On oxytocin drip rate 40 mL/hr
	18.09	COVID-19	COVID-19 RT-PCR test was reported COVID-19 RT-PCR: undetectable
	19.00		PV: Cx 3 cm, 75%, -1, MR FHR 150 bpm, UC 3'10", 45", 2+ On oxytocin drip rate 50 mL/hr
	20.00		FHR 140 bpm, UC 2'50", 40", 2+ On oxytocin drip rate 50 mL/hr
	20.20-20.40	Intrauterine resuscitation	FHR 140 bpm, early deceleration, UC 1"-2'30", 15"-30", 2+ Notify OB-GYN resident 2 Intrauterine resuscitation
	21.00		PV: Cx 5 cm, 80%, 0, MR FHR 150 bpm, UC 3', 45", 2+ On oxytocin drip rate 50 mL/hr
	22.00	Intrauterine resuscitation	V/S BT 37.3 C, P 102 bpm, R 20/min, BP 136/73 mmHg FHR 150 bpm, variable deceleration, UC 2'50", 45", 2+ Hold oxytocin
	22.10		PV: Cx 7 cm, 100%, 0, MR
	23.00		FHR 150 bpm, UC 3'50", 45", 2+ On oxytocin drip rate 10 mL/hr
	23.30		PV: Cx 9 cm, 100%, 0, MR
Oct 21	00.30	2 nd stage of labour	PV: Cx 10 cm, 100%, +1, MR
	00.40		FHR 140 bpm, UC 2'40", 45", 2+ On oxytocin drip rate 10 mL/hr
	01.40		FHR 144 bpm, UC 2'50", 45", 2+ On oxytocin drip rate 10 mL/hr

DATE	TIME	EVENT	DESCRIPTION
	02.00		V/S BT 38.9 C, P 116 bpm, R 20/min, BP 131/81 mmHg
	02.21		Notify OB-GYN resident 2
	02.25		Notify OB-GYN resident 3
	02.30	Prolonged 2 nd stage	
	02.39		PV: Cx 10 cm, 100%, +2, MR Notify OB-GYN staff for vacuum extraction permission Consult and ask the patient for vacuum extraction permission
	03.05-03.12	Active management for 3 rd stage of labour	03.05-03.12 Vacuum extraction 03.12 Fetal delivery, Apgar 6, 8, 8 03.12 Oxytocin 10 units IM
	03.15	Haemorrhagic decompensatory shock	V/S P 150 bpm, BP 86/51 mmHg
	03.16-04.10	PPH Active management for 3 rd stage of labour	Controlled cord traction 03.16 Placental delivery 03.16 Perineal 2 nd degree tear, uterine atony, EBL 500 mL 03.16-04.10 Episiotomy wound and perineal tear were sutured 04.10 Good uterine tone PV: good perineal wound (no bleeding, no hematoma) EBL 1,000 mL Uterine massage
	03.20	Uterotonic drugs Resuscitation	Methylergometrine 200 ug IV NSS 500 mL IV load Retain Foley catheter
	03.25	Uterotonic drugs	NSS 1,000 mL + oxytocin 80 units IV drip rate 120 mL/hr
	03.30		V/S P 152 bpm, BP 84/50 mmHg
	03.45		V/S P 92 bpm, BP 106/85 mmHg
	03.50	Resuscitation	NSS 500 mL IV load (total IV fluid 1,000 mL)
	03.56	X-Match	X-match for PRC was collected

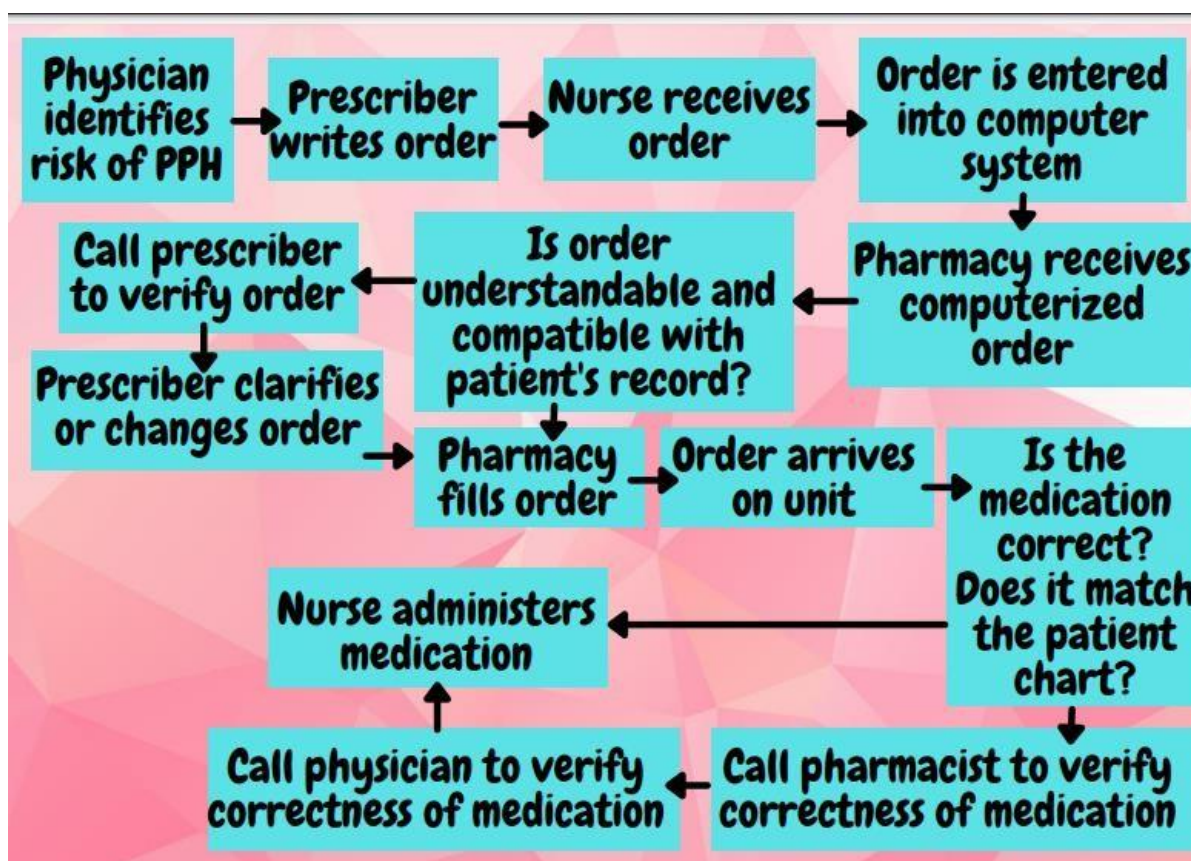
DATE	TIME	EVENT	DESCRIPTION
			Blood specimen was collected
	04.00		V/S P 78 bpm, BP 117/61 mmHg
	04.30		V/S P 117 bpm, BP 95/63 mmHg
	04.43		Blood specimen was taken at Lab
	04.56	X-Match	Specimen was taken at Lab
	05.00		V/S P 110 bpm, BP 108/67 mmHg Urine output 50 mL (for 1 hr 40 min)
	05.03		Coagulogram result was reported PT 13.4 sec, aPTT 22.6 sec, INR 1.20
	05.10		CBC result was reported Hb 9.5 g/dL, Hct 28.1 %, WBC 31,190 cells/uL (N 92%, L 4%), Plt 339,000 /uL
	05.15	Haemorrhagic shock	V/S P 180 bpm, BP 59/32 mmHg
	05.30		V/S P 180 bpm, BP can't be evaluated E ₄ V ₅ M ₆ , drowsiness Notify OB-GYN resident 2 On O ₂ cannula
	05.32-05.35		Notify OB-GYN resident 3 Call blood bank for emergency PRC Good uterine tone PV: good perineal wound (no bleeding, no hematoma)
	05.37	X-Match	V/S P 182 bpm, BP 59/38 mmHg PRC were delivered
	05.40	Resuscitation	LRS 1,000 mL IV load
	05.43		Notify OB-GYN staff Consult anaesthesiologist for central line insertion
	05.45	X-Match	PRC were arrived
	06.00-06.59	Resuscitation	V/S BT 38.1 C, P 164 bpm, R 22/min, BP 137/114 mmHg PRC 5 units, LRS 1,000 mL IV push 06.59 E ₄ V ₅ M ₆ , good consciousness, well cooperate

DATE	TIME	EVENT	DESCRIPTION
	06.06		V/S P 150 bpm, BP 84/62 mmHg
	06.10		Central line insertion
	06.21		V/S P 119 bpm, BP 127/103 mmHg
	06.50		V/S P 124 bpm, BP 150/83 mmHg
	07.50	Clinical improved	V/S P 121 bpm, BP 134/92 mmHg
	08.50		V/S P 130 bpm, BP 133/88 mmHg Urine output 250 mL (for 5 hr 30 min)
	09.05	Empirical antibiotics	Cefoxitin 1 gram IV every 6 hrs
	09.15	Clinical improved	V/S BT 38.8 C, P 120 bpm, R 20/min, BP 130/75 mmHg
	11.00		E ₄ V ₅ M ₆ , good consciousness, well cooperate Good uterine tone PV: no bleeding on sanitary pad, good perineal wound (no bleeding, no hematoma) Start oral diet Iron supplement
	11.37		Blood specimen was collected
	12.00	Clinical improved	V/S P 113 bpm, BP 120/64 mmHg
	12.18		Blood specimen was taken at Lab
	13.30	Clinical improved	V/S P 108 bpm, BP 115/62 mmHg
	13.32		CBC and coagulogram result was reported Hb 10.9 g/dL, Hct 32.0 %, WBC 25,370 cells/uL (N 84%, L 9%), Plt 165,000 /uL PT 14.4 sec, aPTT 25.1 sec, INR 1.29
	14.00		V/S BT 36.6 C, P 108 bpm, R 20/min, BP 108/63 mmHg
	15.00		V/S P 119 bpm, BP 112/63 mmHg
	18.00		V/S BT 36.8 C, P 100 bpm, R 20/min, BP 108/90 mmHg
	22.00		V/S BT 36.7 C, P 100 bpm, R 20/min, BP 119/69 mmHg

DATE	TIME	EVENT	DESCRIPTION
Oct 21	And later	Clinical stable	V/S BT 36.5-37.1 C, P 78-116 bpm, R 20/min, BP 108-135/62-83 mmHg E ₄ V ₅ M ₆ , good consciousness, well cooperate No further bleeding per vagina

PPH Prevention and Flow Chart

- ✓ Uterotonic drugs
 - ✓ Oxytocin 10 units IV bolus or IV infusion or IM
 - Carbetocin 100 ug IV or IM
 - Ergometrine 0.2 mg IV or IM
 - Misoprostol 400-600 ug oral
- ✓ Controlled cord traction
- Late cord clamping
 - ✓ Not recommended in neonatal asphyxia
- ✓ Evaluate uterine tone



PPH Patient Safety Checklist and Flow chart

Acute PPH from vaginal delivery 500-1,000 mL

- ✓ Notify OB-GYN
- ✓ NSS 1,000 mL IV rate 150-200 mL/hr
- In case of continuation bleeding, increase IV rate to 200 mL in 10 min
- ✓ NSS 1,000 mL + oxytocin 40-80 units IV rate 120 mL/hr
- Transamine 1g + NSS 100 mL IV in 30 min
 - Reevaluate next 30 min; if not improved, transamine 1g + NSS 100 mL IV in 30 min
- ✓ Retain Foley catheter
- ✓ Uterine massage
- ✓ In case of uterine atony, methylergometrine 0.2 mg IM every 2-4 hrs
- ✓ X-match for PRC 2 units
- ✓ Evaluate cause of acute PPH e.g., retained product of conception, lacerations, uterine atony and uterine inversion
- Cytotec 800-1,000 ug rectal suppo or Nalador 500 ug + NSS 500 mL IV rate 100 mL/hr

Acute PPH from vaginal delivery 1,000-1,500 mL

- ✓ Notify 2nd call OB-GYN, anaesthesiologist (in case requiring circulation support)
- ✓ Blood for CBC, BUN/Cr, LFT, coagulogram, fibrinogen level
- ✓ Blood component (PRC, FFP, platelet) transfusion depending on patient status
- IV access with 18-gauge needle for NSS 1,000 mL IV 150-200 mL/hr (load 1-2 L IV in 1 hr)
- ✓ O₂ supplement
- In case requiring OB operations (uterine curettage, suture perineal wound), move the patient to OR
- Consider intrauterine balloon or uterine packing
- Warm blanket or warmer to prevent hypothermia
- ✓ X-match for more PRC 2-4 units and FFP 2-4 units

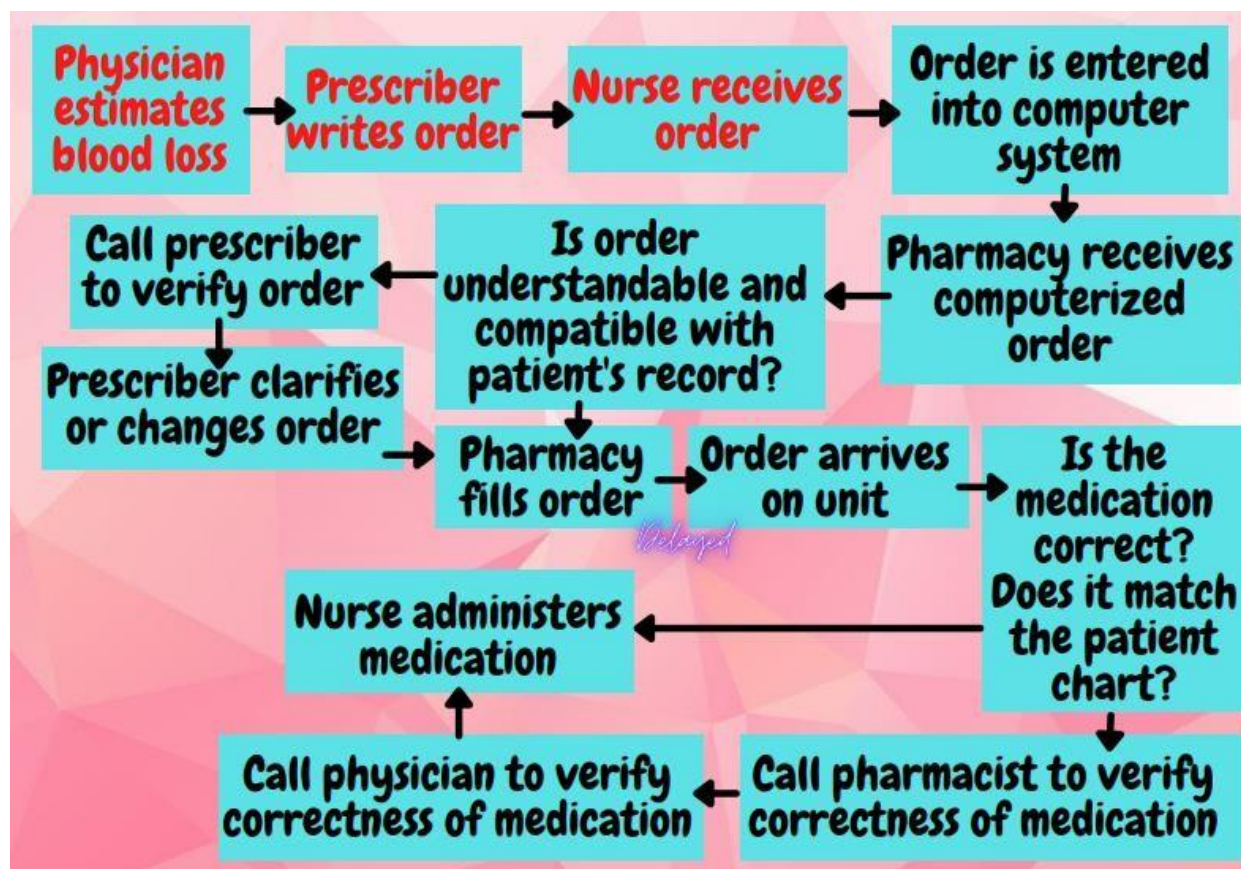
Acute PPH from vaginal delivery more than 1,500 mL

- Massive transfusion protocol
- PRC:FFP:platelet 4:4:1
- Consider uterine artery ligation, B-Lynch sutures or hysterectomy

Haemorrhagic shock resuscitation and Flow Chart

Classification of haemorrhagic shock

Parameters	Compensated	Mild	Moderate	Severe
Blood loss	< 1,000 mL	1,000-1,500 mL	1,500-2,000 mL	> 2,000 mL
HR	< 100 bpm	100-120 bpm	120-140 bpm	> 140 bpm
BP	Normal	Orthostatic	Marked fall (sBP < 90 mmHg)	Profound fall (sBP < 60 mmHg)
Capillary refill	Normal	May be delayed	Usually delayed	Always delayed
Urine output	> 30 mL	20-30 mL	5-20 mL	Anuria



Root Cause Analysis

1. Delayed diagnosis of haemorrhagic shock due to inaccurately estimated blood loss
 - Volume of blood loss should be estimated by BRASSS-V blood collection drape, not visual estimation.
 - In the case of BRASSS-V blood collection drape not providing an accurate volume of blood loss, parameters in the classification of haemorrhagic shock should be used.
2. Two-sited IV access should be performed in case of risk of PPH and obesity.
3. Proper pre-operative evaluation for central line insertion